Supplement: Exposure-Controlled Data Minimization for Personalized LLM Prompts

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1 Scope and interpretation

All disclosure measurements are relative to Qwen and Gemma auditors, not human-validated privacy or harm rates. The bound concerns expected profile exposure for an exchangeable user. Counting the always-sent query as one record yields $E_{\text{total}} = E_{\text{profile}} + a(q)$ and hence an expected bound of $\alpha+1$, because $0 \le a(q) \le 1$; query disclosure prevalence was not measured. A query-only prompt is not a zero-loss total-prompt fallback.

2 Additional readers and rating-only controls

Table 1 gives the 14B results moved out of the main table. The 14B run uses 300 users; the main 7B run uses 450. Comparisons across readers are therefore not paired estimates of a model-size effect. All significance marks and corresponding loss/gain claims use two-sided 90% user-bootstrap intervals. Non-inferiority requires the lower 90% bound of the retained-gain ratio to exceed 75%. Descriptive effect-size intervals use 95%; exploratory rule-by-target comparisons are unadjusted.

Table 1: LaMP-3 with Qwen2.5-14B: exposure, admitted records and utility. Marks have the same definitions as Table 1 of the paper.

	LaMP-3, 14B		
Admission rule	Exp.	Rec.	Util.
Uncalibrated, whole records			
No profile	0.00	0.0	0.863
Similarity top-5	0.90	5.0	0.928
Detector filter top-5	0.53	4.0	0.916
Uncalibrated, rating summaries or text removal			
Ratings only	0.00	5.0	0.909^{-}
Redact detector-flagged text	0.42	5.0	$0.932^{\dagger}$
Redact auditor-flagged text	0.00	5.0	$0.930^{\dagger}$
Union calibration: CRC, $\alpha = 0.2$			
Similarity top-k	0.09	0.5	0.888^{-}
Similarity threshold	0.18	1.3	0.892^{-}
Detector filter	0.10	1.0	0.900^{-}
Cleared similarity	0.14	3.2	0.907^{-}
Union calibration: LTT, $\alpha = 0.2$, $\delta = 0.1$			
Similarity top-k	0.00	0.0	0.863^{-}
Similarity threshold	0.09	0.5	0.884^{-}
Detector filter	0.00	0.0	0.865^{-}
Cleared similarity	0.05	1.8	0.897^{-}

The mean-rating control sends the arithmetic mean of *all* available past profile ratings, rounded to four decimal places, followed by the unchanged target review and rating instruction. It sends no past review text or individual past ratings. The reader, backend, temperature, seed and two-token generation limit match the

main 7B evaluation. The statistic is still personal data: its zero record-text count is not a privacy guarantee. Paired bootstrap resampling uses 2,000 user samples and seed 20261005. The mean-only utility is 0.893 on 450 users, versus 0.883 without a profile and 0.922 with top-5. It retains 26.8% of the gain (90% interval 1.3–48.4%); its difference from top-5 is -0.029 [-0.041, -0.017] (90%).

3 Independent-family judge and fresh auditor repeats

Gemma-2-9B rescored all 3184 saved main-reader answers for 150 LaMP-QA users, including every admission context. Top-5 minus no-profile utility is -0.006 [-0.031, +0.022] under Gemma, versus +0.016 [-0.005, +0.040] under Qwen (95% paired user-bootstrap intervals; 2,000 shared resamples, seed 20261005). This tests model-rubric sensitivity, not human correctness.

Gemma returned 3184/3184 complete valid score vectors; 0 replies were malformed, 0 had the wrong length, and 0 required value coercion or clipping. 0 original replies were unavailable for this parsing check.

Qwen returned 3043/3184 complete valid score vectors; 0 replies were malformed, 101 had the wrong length, and 42 required value coercion or clipping. 0 original replies were unavailable for this parsing check.

These diagnostic categories can overlap. The original scoring parser is retained; the independent Gemma results therefore provide a check with complete valid rubric vectors.

Rule	α	Qwen judge	Gemma judge
No profile	–	0.608	0.454
Top-5	–	0.624	0.448
Similarity top-k	0.2	0.608	0.454
Similarity threshold	0.2	0.609	0.457
Detector filter	0.2	0.607	0.452
Cleared similarity	0.2	0.628	0.455
Similarity top-k	0.5	0.606^{-}	0.434
Similarity threshold	0.5	0.610	0.457
Detector filter	0.5	0.598^{-}	0.436
Cleared similarity	0.5	0.614	0.439

Cross-fitted utility with policies, disclosure labels, grids and splits held fixed. Marks compare each rule with top-5 under the same judge using 90% intervals. Changing the judge does not refit the gate or change the exposure calibration.

Fresh qwen2.5:7b calls agree on 96/96 binary flags and 96/96 complete category sets with the original outputs (32 uniformly sampled top-eight evaluation records per task; seed 20261005). The repeat cache is separate and does not read the original cache; interrupted repeats resume only their own calls.

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Agreement on this sample is not proof of determinism on all records and does not measure label validity.

4 LaMP-2 pilot

A separate 60-user Qwen2.5-14B pilot increased accuracy from 0.333 to 0.517, a gain of +0.183 [+0.067, +0.300] (95% interval). This passed the declared go criterion (gain at least 0.05 with a positive lower bound). The 150-user extension is incomplete; this is preliminary task-selection evidence, not independent confirmation. The endpoint-backed pilot is not silently combined with a different backend. A separate 40-user LaMP-QA pilot with the 14B reader had gain -0.022 [-0.070, +0.026] under the Qwen rubric judge (95% interval); it is also preliminary.

5 Severity sensitivity and repeated disclosures

The exploratory severity loss assigns 2 to a record with any health, mental-health, sexual-orientation, race/-nationality or finance union flag, 1 to another flagged record, and 0 otherwise. Overlapping categories use

the maximum weight. With a five-record cap the loss bound is 10. Weights are illustrative policy choices, not a validated harm scale or legal classification. At the same numerical target the units differ from uniform counts. The table uses 300 matched random calibration/test halves, the original cleared-rule grid and union labels; numbers are average held-out losses, not confidence bounds. No utility conclusion follows from these admission counts.

Task	Count loss	Records	Weighted loss	Records
LaMP-QA	0.107	2.18	0.086	1.28
LaMP-3	0.150	3.11	0.090	1.98
LaMP-7	0.000	0.00	0.000	0.00

Real multi-query cumulative distinct disclosures are not measured. The artifact contains a limited synthetic sensitivity: independently shuffled rankings of the same audited top-eight pool for up to ten arrivals, with the original query-specific scores fixed. On LaMP-3 at $\alpha = 0.5$, single-prompt calibration produces 0.375 unique flagged records at one arrival and 0.490 at ten; calibrating the cumulative loss directly gives 0.351 at ten. There are no new queries, retrieval beyond the fixed pool, or measured utility. These results do not establish safety for adaptive interaction. Linearity of expectation gives a $T\alpha$ bound on record occurrences only when each prompt's marginal assumptions remain valid; distinct records are at most those occurrences, with no individual-user guarantee.

6 Human audit status

No human annotations have been completed. Two independent blind annotators and human adjudication are required before reporting precision/recall or a human-calibrated operating point. The existing 120-record stratified packet has Gemma coverage on only 96 records and contains no fully labelled user pool. Its descriptive metrics cannot identify an exact operating-point shift. Missing labels are bounded, never treated as negative labels. Full top-eight pools from disjoint calibration and test users are required; scores and the threshold grid must remain frozen. The private annotation workflow checks category consistency, independence attestations and adjudication. Model repeatability below does not replace human validation. A second prepared packet covers all eight candidates of 60 uniformly sampled LaMP-3 users (480 records per annotator), with 30 calibration and 30 disjoint test users fixed before annotation. Rule and grid hashes prevent changes during this study. It can identify a small-cohort operating-point shift after genuine annotation; it currently contains zero labels and provides no human result.

7 Auditor and judge reproducibility

Each auditor receives one displayed record per call. Record text is truncated at 60 words for LaMP-3, 40 for LaMP-7 and 90 for LaMP-QA; LaMP-3 retains its rating prefix. Both auditors use greedy decoding (temperature 0), seed 0, a 40-token output limit and JSON output. Qwen is served by Ollama 0.30.8; Gemma uses its bundled llama.cpp server with the same local GGUF weights, `--jinja`, an 8192-token total context and four slots. The original parser filters category names and maps malformed replies to an empty list; repeat agreement therefore measures the complete model-and-parser pipeline. The second judge uses Gemma with the same rubric prompt and 40-token limit as the Qwen judge. Scores are clamped to 0–2, truncated to the rubric length, missing entries padded with zero, and malformed replies scored zero. The following SHA-256 digests were verified against the local GGUF bytes on 2026-10-10.

qwen2.5:7b Manifest SHA-256:

`845dbda0ea48ed749caafd9e6037047aa19acfcfd82e704d7ca97d631a0b697e`

GGUF SHA-256:

`2bada8a7450677000f678be90653b85d364de7db25eb5ea54136ada5f3933730`

gemma2:9b Manifest SHA-256:
`ff02c3702f322b9e075e9568332d96c0a7028002f1a5a056e0a6784320a4db0b`
GGUF SHA-256:
`ff1d1fc78170d787ee1201778e2dd65ea211654ca5fb7d69b5a2e7b123a50373`

Exact auditor prompt template

The placeholder {`text`} is the displayed record. Doubled braces in these Python format templates escape literal JSON braces.

```
You are a privacy auditor. You detect SELF-DISCLOSURE: statements in which the author reveals personal information about
     themselves or about a specific person close to them.

Categories:
AGE_GENDER: the author's age or gender
RACE_NATIONALITY: the author's race, ethnicity or nationality
LOCATION: where the author lives, works, studies or is travelling (city, region, country, address)
APPEARANCE: the author's body, height, weight or looks
PARTNER: the author's wife, husband, girlfriend, boyfriend, or relationship status
FINANCE: the author's income, debts, purchases they can or cannot afford, financial situation
OCCUPATION: the author's job, employer, profession or workplace situation
FAMILY: information about the author's children, parents, siblings or other relatives
HEALTH: physical health conditions, injuries, medication, pregnancy, disability
MENTAL_HEALTH: mental health conditions, therapy, emotional struggles
EDUCATION: the author's school, degree, studies or exams
PET: the author's own pets
SEXUAL_ORIENTATION: the author's sexual orientation
NAME_CONTACT: real names of private persons, phone numbers, emails, private account handles

A text counts only if it reveals a fact about the author's own life or a person close to them. Opinions about products, films,
      news or public figures, general questions, and generic statements reveal nothing and get an empty list.

Text: """{text}"""

Reply with JSON only: {{"categories": [category names disclosed in the text, empty list if none]}}
```

Exact judge prompt template

Each aspect is its numbered name and reason truncated to 40 words.

```
You grade an answer to a user's question against the aspects that this user needs addressed.

Question: {q}

Answer: {a}

Aspects:
{aspects}

For each aspect give a score: 0 = not addressed, 1 = partially addressed, 2 = fully addressed.
Reply with JSON only: {{"scores": [one integer per aspect, in order]}}
```